

4 Cancers can develop as a result of mutations or epigenetic modifications.

(a) (i) State what is meant by the term **mutation**.

(1)

(ii) Name the type of nuclear division taking place as a tumour develops.

(1)

(b) How many of the following statements about epigenetic modification are correct?

(1)

- It can result in a change in the base sequence of a gene
- It can involve histone modification
- It can result in an altered phenotype
- It can involve DNA methylation

☐ **A** 1

☐ **B** 2

☐ **C** 3

☐ **D** 4

(c) Oestrogen stimulates some types of breast cancer to proliferate.

Oestrogen is a steroid hormone found in mammals.

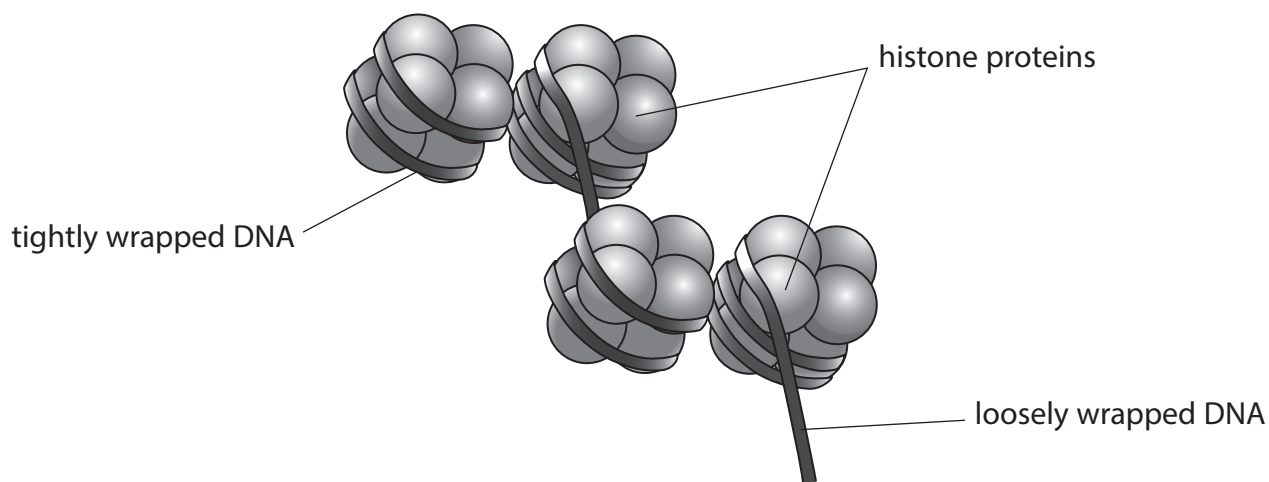
Oestrogen affects up to 100 different genes.

Explain how oestrogen may result in the proliferation of some breast cancer cells.

(4)

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- (d) The expression of genes in cancer cells can be regulated by histone modification.
- DNA that is not being transcribed is tightly wrapped around histone proteins.
- The diagram shows DNA wrapped around histone proteins.



Acetylation of histones is one example of histone modification. Acetyl groups are added to histone proteins.

Suggest how acetylation of histone proteins could result in increased gene expression.

Use the information in the diagram to support your answer.

(2)

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(Total for Question 4 = 9 marks)